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Transformer - I



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Applications and Demos of Transformer Models

Real-world Uses Of Sequence Models

Real-world Applications

Sequence-to-sequence models in production

Video Captioning

Turning visual sequences into textual descriptions

Text Summarisation

Transformer-powered content compression

Machine Translation Demo

Hands-on transformer-based translation

Attention Applications

Use cases across different industries

Industry Case Studies

Examples from healthcare, finance, and retail

Quick Refresher: Core Concepts

Revisiting the Building Blocks

Key Components Review

Encoder-Decoder

Transforms input sequences

Attention Mechanism

Focuses on key information

Transformers

Enable parallel processing

Today's Focus: From theory to practice applying these concepts to real-world problems

Neural Machine Translation in Action

Real-world Use of Transformer Models

Google Translate

Handles 100+ billion words daily across 108 languages using transformers for real-time translation

DeepL

Uses attention mechanisms for near human-level accuracy, especially in European languages

Key Advantage: Parallel processing cuts translation time and reduces training from weeks to days

Demo: Machine Translation with MarianMT

Live Translation Example

```
from transformers import pipeline

# Initialize translation pipeline
translator = pipeline(
    "translation_en_to_fr",
    model="Helsinki-NLP/opus-mt-en-fr"
)

# Input text
text = "Machine learning transforms how we process language."

# Generate translation
result = translator(text)

# Print result
print(result[0]['translation_text'])
```

English: “Machine learning transforms how we process language”

French: “L'apprentissage automatique transforme notre façon de traiter le langage »

Key Point: The model understands the meaning of the text, not just individual words. Its attention mechanism ensures context is maintained even when translating between languages

Video Captioning: From Sequences to Language

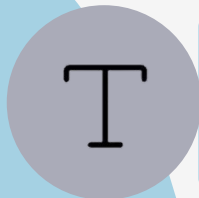
Turning Visual Frames Into Meaningful Text



Video Frames: Visual inputs captured at set intervals



Feature Encoding: CNN extracts features, transformer analyses the sequence

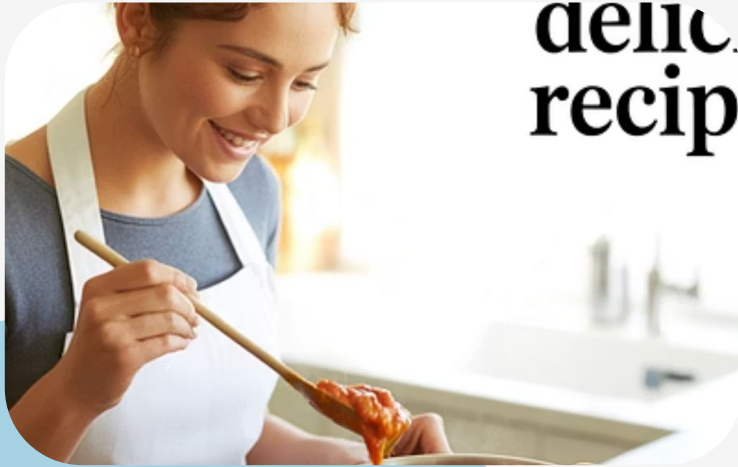


Caption Generation: Description of video content in natural language

Real-world Application: YouTube's auto-captioning make videos accessible and searchable through NLP

Demo: Automated Video Captioning

Understanding Scenes Through AI



Generated Caption: "A person chops vegetables in a modern kitchen while explaining cooking techniques"



Generated Caption: "A herd of elephants walks across the African savanna during golden hour"



Generated Caption: "A basketball player makes a three-point shot as the crowd cheers"

These captions show how the model recognises actions, objects, and context within video frames

Text Summarisation with Transformers

AI Models That Condense Information



News: Reuters and AP News use BART models to summarise articles, saving time while keeping key details



Research: Semantic Scholar applies Pegasus transformer to summarise papers, helping researchers find relevant studies faster



Meetings: Microsoft Teams uses T5 models to generate action items and key points from meeting recordings

Demo: Abstractive vs Extractive Summarisation

How Transformers Create Concise Summaries

Original Text:

"Transformer models have revolutionized natural language processing by introducing the attention mechanism, which allows models to focus on different parts of the input sequence when generating each output token. Unlike recurrent neural networks, transformers process all positions in parallel, leading to significant speedups during training. The self-attention mechanism computes attention weights for each token with respect to all other tokens in the sequence..."

Abstractive Summary:

"Transformers use attention mechanisms for parallel processing, replacing RNNs with faster training. Self-attention enables dynamic focus on relevant sequence parts, revolutionizing NLP through improved efficiency and performance."

Here, the model rephrases the text while keeping the original meaning — this is abstractive summarisation in action.

Attention Mechanisms Across Industries

Real-World Applications of Attention Models



Healthcare:

Mayo Clinic uses attention models to summarise clinical notes, cutting doctor review time by 40%.



Finance:

JPMorgan applies sequence attention to detect fraud by analysing transaction patterns with 95% accuracy.



Retail:

Amazon uses multi-head attention to track browsing behaviour, improving recommendations and boosting conversions by 23%.

Sentiment Analysis for Customer Service

Using Transformers to Classify Customer Feedback

```
from transformers import pipeline

# Initialize sentiment classifier
classifier = pipeline(
    "sentiment-analysis",
    model="cardiffnlp/twitter-roberta-base-sentiment-latest"
)

# Customer reviews
reviews = [
    "The product arrived quickly and exceeded my expectations!",
    "Terrible customer service, waited 3 hours on hold.",
    "Average product, decent price, would consider buying again."
]

# Analyze sentiment
for review in reviews:
    result = classifier(review)
    print(f"Review: {review}")
    print(f"Sentiment: {result[0]['label']} ({result[0]['score']:.3f})")
```

Sentiment Analysis for Customer Service

Using Transformers to Classify Customer Feedback



Positive Detection: Accuracy on customer satisfaction



Negative Detection: Identifying service issues



Neutral Recognition: Balanced feedback classification

Multi-Head Attention

How Models Capture Word Relationships

1. **Input Processing:** Each word generates query, key, and value vectors
2. **Attention Computation:** Multiple heads focus on different word relationships
3. **Weighted Combination:** Attention weights decide each word's influence
4. **Output Generation:** Produces context-aware representations for the next layer



Interactive Tool: BertViz visualises attention patterns in real time, showing how models understand language dependencies.

Key Takeaways & Applications

The Broad Impact of Transformer Models

Scalable Architecture:

Transformers support parallel processing, enabling large-scale sequence tasks across languages and domains

Improved Interpretability:

Attention mechanisms make model decisions more transparent, essential in healthcare, finance, and regulated sectors

Cross-Industry Impact:

Used in translation, fraud detection, and more, transformers show the wide value of attention-based models

Q&A Session



Open floor for questions and clarifications.